Software Requirement Document (SRD)

Inventory Management System

For Siddesh Technologies Pvt. Ltd.

Version: 1.0

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1. Objective

Develop an Inventory Management System to maintain complete records of all inventory movement within the organization.

The system should support

- Product Master
- Barcode Generation
- Barcode Scanning
- Inward Entry
- Outward Entry
- Current Stock
- Inventory History
- Reports

The application should eliminate manual register entries and maintain complete traceability of every product.

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2. User Roles

Initially only one Admin login.

Later support

- Admin
- Store Manager
- Sales Executive

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3. Product Master

Admin should be able to create products.

Fields

- Product Name
- Product Category

- Brand
- Model Number
- Unit
- Product Description
- Minimum Stock Alert
- HSN Code (optional)
- GST Percentage (optional)

Example

AI Lab

- Arduino UNO
- Servo Motor
- RFID Module
- Drone Kit
- Robotics Kit
- VR Headset

Digital Products

- 64 GB Pen Drive
- 128 GB Pen Drive

Office Items

- Mouse
- Keyboard
- HDMI Cable

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4. Barcode Management

Every physical product should have a unique barcode.

Two options

Option A

Generate barcode from software.

Example

ST00000001

ST00000002

ST00000003

Software prints barcode label.

Option B

User pastes manufacturer barcode already available.

Software stores it.

Both methods should be supported.

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5. Inward Process

Workflow

Step 1

Select Product

or

Create New Product

Step 2

Scan Barcode

If barcode not available

Generate barcode.

Step 3

Enter Quantity

Example

20 Pen Drives

Step 4

Enter Supplier Details

Fields

Supplier Name

Supplier Mobile

GST No (optional)

Invoice Number

Invoice Date

Purchase Order Number

Step 5

Enter Person Who Brought Material

Examples

Atharva Birari

Courier

Blue Dart

DTDC

Supplier Representative

Step 6

Upload Invoice (optional)

PDF

Image

Step 7

Click

Save Inward

Software should

Increase Stock

Store complete inward history

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6. Outward Process

Workflow

Step 1

Scan Barcode

or

Search Product

Step 2

System automatically displays

Product Name

Current Stock

Available Quantity

Step 3

Enter Quantity

Step 4

Select Outward Type

- Sale
- Demo
- Replacement
- Internal Use
- Service
- Sample

Step 5

Enter Party Details

School Name

Contact Person

Mobile Number

GST Number (optional)

Address

Invoice Number

Sales Order Number

Step 6

Enter Person Handing Over Material

Step 7

Receiver Name

Who collected material.

Step 8

Optional Signature

Digital Signature

or

Photo Capture

Step 9

Click

Save

Software should reduce stock.

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7. Current Stock

Dashboard showing

Product

Opening Stock

Inward

Outward

Current Stock

Reserved Stock (future)

Available Stock

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8. Barcode Scanner Support

Software should support

USB Barcode Scanner

Bluetooth Barcode Scanner

Camera Scanner (future)

When barcode is scanned

Cursor automatically searches product.

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9. Barcode Label Printing

Support

Standard Barcode Labels

Example

Arduino UNO

ST00012345

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Print multiple labels.

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10. Search

Search by

Barcode

Product Name

Category

Supplier

Invoice Number

School Name

Date

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11. Reports

Inward Report

Date-wise

Supplier-wise

Product-wise

Outward Report

School-wise

Invoice-wise

Date-wise

Sales Executive-wise

Stock Report

Current Stock

Low Stock

Out of Stock

Product Ledger

Shows complete history

Example

01 Jan

Inward

+20

Supplier ABC

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05 Jan

Outward

-2

XYZ School

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10 Jan

Outward

-5

ABC School

Current

13

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12. Dashboard

Cards

Current Stock

Today's Inward

Today's Outward

Low Stock

Pending Orders (future)

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13. Barcode Search Logic

When barcode scanned

If barcode exists

Open Product

Else

Display

Barcode not found.

Create New Product?

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14. Audit Trail

Every transaction should store

Created By

Created Date

Updated By

Updated Date

Computer Name (optional)

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15. Future Ready Modules

Leave architecture ready for

Purchase Orders

Sales Orders

Vendor Management

School Management

AMC Tracking

Warranty Tracking

Repair History

Asset Tracking

QR Code Support

Serial Number Tracking

Multi Warehouse

Multi Branch

Mobile App

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16. Database Tables

Product Master

- Product ID
- Product Name
- Category
- Barcode
- Model
- Brand
- Unit
- Description
- Minimum Stock

Inward

- Inward ID
- Product ID
- Barcode
- Quantity
- Supplier
- Invoice No
- Invoice Date
- Purchase Order
- Received By
- Entered By
- Date

Outward

- Outward ID
- Product ID
- Barcode
- Quantity
- Party

- Invoice No
- Outward Type
- Delivered By
- Received By
- Date

Stock Ledger

Every stock movement.

Never delete.

Only append.

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17. Technical Requirements

- Desktop application (Electron + React, matching your existing AI Lab LMS technology stack)
- SQLite for single-office deployment (upgrade path to PostgreSQL/MySQL for multi-user or cloud deployments)
- USB barcode scanner support (keyboard-emulation scanners)
- Barcode generation (Code 128 preferred)
- Barcode label printing
- PDF report export
- Excel export
- Daily automatic database backup
- Restore from backup
- User authentication with role-based access (future-ready)

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18. Suggested Enhancements for Your Business (Highly Recommended)

Since your business deals with AI Lab kits made up of multiple components, I recommend adding these features from the beginning:

A. Kit/Bundle Management (Bill of Materials)

An **AI Lab Kit** should be treated as a bundle containing multiple items.

Example:

- 1 × Arduino UNO

- 1 × Servo Motor
- 1 × Ultrasonic Sensor
- 1 × RFID Module
- 1 × Breadboard
- 20 × Jumper Wires

When **one AI Lab Kit** is sold, the system should automatically deduct the quantities of all constituent components from inventory.

B. Serial Number Tracking

Certain products (VR headsets, laptops, drones, tablets, printers, monitors, etc.) should have individual serial numbers. Each serial number can have its own barcode, making warranty claims, replacements, and servicing much easier.

C. Pen Drive Content Tracking

For your educational content business, maintain separate records for:

- Pen Drive Capacity (32 GB / 64 GB / 128 GB)
- Content Version (e.g., Std. 1–4 Marathi v2.3)
- Content Loaded Date
- Loaded By
- Write Protection Status (if applicable)

This lets you identify exactly which content version was supplied to a school.

D. School Asset History

For every school, maintain a complete transaction history:

- AI Lab Kits supplied
- Pen drives supplied
- Replacements
- Additional purchases
- Warranty/service records

This creates a complete customer asset ledger.

E. Stock Movement Timeline

Each product should have a chronological timeline showing:

- Date and time
- Inward/Outward action
- Quantity

- User who performed the action
- Supplier or customer
- Balance stock after the transaction

This is invaluable for audits and troubleshooting.